R&D MANAGEMENT

In search for the not-invented-here syndrome: the role of knowledge sources and firm success

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The not-invented-here (NIH) syndrome refers to a negative attitude of employees against externally developed knowledge. We show that the sources of external knowledge and the success of the company that acquires knowledge externally are factors that impact the occurrence of the NIH syndrome. In line with social identity theory, we hypothesize that internal resistance is most likely to occur if knowledge is acquired from similar organizations. This hypothesis is supported by our empirical finding that internal resistance against external knowledge is more likely to occur when knowledge is acquired from competitors rather than from suppliers, customers, or universities. Further, we show that the NIH syndrome is more likely to arise within successful companies that acquire knowledge from competitors. This is in line with our hypothesis that firm success increases the extent to which employees identify themselves with their company, resulting in stronger in-group favoritism and a superior tendency to reject externally generated knowledge.

1. Introduction

Innovation management has to pay careful attention to the fact that the institutional locus of technological advances can lie beyond the firm's boundaries (Teece, 1986, 1992). A skillful combination of externally acquired knowledge with the firm's own knowledge base can have substantial effects on firm performance and competitiveness (e.g. Kogut and Zander, 1992; Teece, 1986, 1992; Laursen and Salter, 2006; Chesbrough, 2003; Rosenkopf and Nerkar, 2001; Cassiman and Veugelers, 2006; Huizingh, 2011; Lichtenthaler, 2011).

Successful adaption and implementation of acquired technologies relies, however, crucially on the openness of the individual employees towards externally developed technologies (Clagett, 1967; Lichtenthaler and Ernst, 2006). A welcoming attitude of employees towards new ideas cannot be taken for granted though (Clagett, 1967; Katz and Allen, 1982). This is because individuals embedded in knowledge creation processes develop routines, beliefs, artifacts, and habits in order to cope with complexities (Dosi, 1982; Nelson and Winter, 1982; Garud and Rappa, 1994; Szulanski, 1999; Garud and Karnoe, 2001; Teece et al., 1997). Routines reinforce path dependencies and limit the rate of integration of external knowledge and the production of radical innovations (Leonard-Barton, 1992; Tripsas, 1997). The assimilation of externally generated knowledge requires individuals to change beliefs, to look beyond the boundaries of their communities, and to break with routines.

Changing beliefs and breaking routines can be a challenge. Within their company, individuals strive for self-enhancement, so they tend to favor their company, their 'in-group', over other companies and aim for a positive distinction from other companies, their 'out-groups' (Ashforth and Mael, 1989; Bartel, 2001). The acceptance of externally generated technologies enforces a comparison of the in-group's technological expertise with that of the out-group. This can constitute a threat for the perceived expertise of a group and, hence, for the self-concept of the group and its members. Employees, working teams, and communities can respond to this threat with resistance towards externally developed knowledge. A negative attitude towards externally developed knowledge is referred to as the not-invented-here (NIH) syndrome (Clagett, 1967; Katz and Allen, 1982).

Although an often discussed phenomenon among practitioners, the NIH syndrome has received relatively little attention in the academic literature (see Antons and Piller, 2015, for a recent survey). The focus of previous studies is on the antecedents of the NIH syndrome such as group tenure (Katz and Allen, 1982), the lack of or negative group experience with external knowledge (Mehrwald, 1999), dysfunctional intra-organizational communication (Mehrwald, 1999) or inappropriate incentive systems (De Pay, 1989, 1995a, 1995b; Mehrwald, 1999).

This study contributes to our understanding of the NIH syndrome by arguing that not only project-related and team-related factors have an impact on the occurrence of the NIH syndrome. We provide a theoretical framework that suggests that the source of externally generated knowledge and the success of the knowledge sourcing company are important antecedents for the NIH syndrome as well. Building on social identity theory (Tajfel, 1974, 1978, 1982; Tajfel and Turner, 1979, 1986; Turner et al., 1987) and the concept of organizational identity (Ashforth and Mael, 1989; Dutton et al., 1994), we suggest that a rejection of external knowledge is most likely if the out-group from which the knowledge is acquired is similar to the in-group. If the out-group shares characteristics important for in-group identification, like expertise in the same technology field or product market, individuals tend to fear their group identity threatened. In-group favoritism and a hostile behavior towards external knowledge can emerge as a defensive mechanism to restore group identity (Gabarott et al., 2009). Among different knowledge sources, competitors are the most similar out-group for companies in terms of product market and technology market expertise as compared with suppliers, customers, and universities. We empirically show that internal resistance against external knowledge, as an indication of the NIH syndrome, is likely to occur when external knowledge is acquired from competitors.

Our theoretical framework further predicts a relationship between a firm's success and the occurrence of the NIH syndrome. The extent to which individuals identify themselves with their company increases with group success because success increases the group's distinctiveness and attractiveness (Blanchard et al., 1975; Dutton et al., 1994). A high confidence in the in-group's capabilities is often accompanied by an increased readiness to degrade outsiders' competencies and, hence, a decreased readiness to accept external knowledge (Katz and Allen, 1982). Because successful groups identify more strongly with their organization and are, hence, more likely to take defensive action against identity threats, we argue and show that the most successful firms are most likely to experience the NIH syndrome when knowledge is acquired from similar sources, i.e. from competitors. The findings of our study should be of interest for managers that organize and optimize the innovation activities of companies. It is important for these decision makers to be aware of potential antecedents of the NIH syndrome in order to be able to establish the means to prevent the NIH syndrome.

The remainder of the paper is organized as follows. The next section presents a review of the literature on the NIH syndrome. Section 3 develops a theoretical framework and derives hypotheses. Section 4 introduces our data set, and section 5 shows the empirical results. The last section concludes.

2. The NIH syndrome: a literature review

In the seminal study on the NIH syndrome, Clagett (1967) analyzes several cases of successful and unsuccessful implementation of process innovations reporting notable resistance against externally developed knowledge. Clagett (1967) emphasizes the importance of individuals in order to prevent the NIH syndrome. Katz and Allen (1982) and Mehrwald (1999) stress the importance of group characteristics such as project team tenure and team experience with external knowledge.[1] De Pay (1989, 1995a, 1995b) shows that miscommunication within an organization and inappropriate incentive systems can be further antecedents of the NIH syndrome. Burcharth et al. (2014) adds that nurturing special talents can lead to the NIH syndrome.

Lichtenthaler and Ernst (2006) and Antons and Piller (2015) provide literature reviews and conceptual frameworks for understanding the NIH syndrome. Lichtenthaler and Ernst (2006) define knowledge management consisting of three 'knowledge management cycles': knowledge acquisition, knowledge accumulation and knowledge exploitation (Argote et al., 2003). At each cycle, not only an excessive negative attitude towards external knowledge but also an excessively positive attitude can occur. They argue that both extremes can be detrimental for the knowledge management within the organization. Antons and Piller (2015) use a different approach distinguishing between different behavioral trajectories of attitudinal functions and relate them to the NIH syndrome.

Although the early studies have focused on the NIH syndrome within individual project teams, some recent studies employed different perspectives on the phenomenon. Agrawal et al. (2010) examine the NIH syndrome employing a geographical focus and a firm-level analysis. They find that especially large companies in the most innovative regions of the United States are susceptible to the NIH syndrome. Companies' tendency to build on own prior inventions disregarding externally developed ideas leads to patens of lower importance. Kathoefer and Leker (2011) shed light on the NIH syndrome in an academic environment concluding that the project experience and the attitude of professors towards science impact the likelihood of the NIH syndrome.

The NIH syndrome, as we perceive it, situates at the knowledge acquisition level of companies. We contribute to the literature by proposing the source of external knowledge and company success as further antecedents of the NIH syndrome.

3. Theoretical framework

3.1. Sources of external knowledge and the NIH syndrome

Knowledge creation is a complex process involving different tasks and individuals with different backgrounds, interests, and information. In order to facilitate the knowledge creation process, organizations develop routines. Individuals and teams develop subroutines within the organizational context in order to support information processing and problem solving (Dosi, 1982; Nelson and Winter, 1982; Szulanski, 1996; Garud and Karnoe, 2001; Van Looy et al., 2001). Such routines evolve over time and are mainly tacit so that they are difficult to be imitated or changed (Teece et al., 1997). They create strong path dependencies regarding the firm's innovation process.[2] Path dependencies support cumulativeness in innovation and facilitate routinized tasks but are not supportive of changes (Cohen and Levinthal, 1990; Leonard-Barton, 1992).

Path dependencies are fostered by community formation and vice versa. Within communities, common beliefs, artifacts, and habits are developed alongside daily activities that create powerful path dependencies (Van Looy et al., 2001). Such path dependencies affect the formation of expectations and the self-concept of individuals and teams (Garud and Rappa, 1994). Individuals identify themselves with their in-group and relate their self-concept and self-esteem to group membership as predicted by social identity theory (Tajfel, 1974, 1978, 1982; Tajfel and Turner, 1979, 1986; Turner et al., 1987) and the concept of organizational identity (Dutton et al., 1994; Ashforth and Mael, 1998). Individuals strive for a positive social identity and engage in self-enhancement within their organization (Tajfel and Turner, 1986; Ashforth and Mael, 1998), which can lead to in-group favoritism (Brewer, 1979; Tajfel and Turner, 1986; Ashforth and Mael, 1989; Abrams and Hogg, 1990). If individuals feel their organizational identity threatened, they show a hostile behavior protecting their organization's self-concept.

External knowledge is a factor that can threaten the self-concept of organizational entities. The confrontation with external knowledge enforces social comparison between the in-group and the knowledge-producing out-group and leads to a re-evaluation of the own organizational identity (Bartel, 2001). The acceptance and valuation of external knowledge can be perceived by insiders as a degradation of the own achievements, expertise, and competence of the in-group. In consequence, individuals may reject external ideas to defend their group identity (Tajfel and Turner, 1979; Brown, 2000). This attitude renders the acceptance, integration, and application of external knowledge difficult or impossible: the NIH syndrome occurs. Hence, our first hypothesis reads:

Hypothesis 1: External knowledge sourcing increases the likelihood of internal resistance against new innovation projects.

As argued above, internal resistance against external knowledge, the NIH syndrome, is consistent with the concept of in-group favoritism. Two important results of social and organizational identity theory suggest that the out-group that generated the externally acquired knowledge should deserve attention when analyzing the occurrence of the NIH syndrome.

First, organizations tend to compare themselves with similar organizations (Ashforth and Mael, 1989; Bartel, 2001). Tensions and the feeling that the in-group's identity is threatened by outsiders intensify with increasing similarity between in-group and out-group (Tajfel, 1974; Tajfel, 1982; Abrams and Hogg, 1990; Branscombe et al., 1993) because similarity between groups increases their comparability (Caddick, 1982) and the boundaries between groups threaten to obliterate (Sanchez-Mazas et al., 1994). Individuals react with increased efforts to reassure distinctiveness and to reinstall the boundaries between groups, which in turn strengthens the in-group bias (Jetten and Spears, 2003).

Second, individuals and organizations are capable of making social comparisons on multiple dimensions. In this sense, organizations can appreciate each other's expertise when they are superior on complementary or distinct dimensions (Ashforth and Mael, 1989). In-group favoritism is strongest on dimensions regarded as important for the in-group, whereas out-group favoritism is likely to occur on dimensions that are less important for the in-group (Mummendey and Schreiber, 1984). In other words, groups are able to acknowledge each other's differential expertise without compromising a positive differentiation. Applying these results to external knowledge acquisitions suggests that the source of external knowledge matters.

3.1.1. Heterogeneity of knowledge sources

Previous innovation literature has acknowledged the heterogeneity of different sources of external knowledge and its contribution to firm performance and innovation (e.g. Belderbos et al., 2004a, 2004b). Prior studies distinguish between knowledge acquired from vertical partners (customer and suppliers), competitors, and universities.

Knowledge from (lead) customers can help defining innovations and reduce the risk associated with market introduction (Von Hippel, 1988; Brown and Eisenhardt, 1995). Innovations triggered by user needs can become a dominant design (Utterback, 1994). In addition, information from customers can support balancing performance and price. Customer knowledge can be, in particular, important for the development of novel and complex new products (Tether, 2002).

Supplier knowledge has been shown to be important for cost reductions within the firms' production process and product quality enhancements (Choi and Hartley, 1996; Ireland et al., 2002; Belderbos et al., 2004b). Information from suppliers can spur fast delivery and decrease production lead time (Choi and Hartley, 1996).

Universities and public research institutions are important sources of science-based knowledge. Scientific knowledge can increase firms' understanding of recent scientific and engineering advances, facilitate the recruitment of R&D personnel, grant access to scientific networks, and reduce costs of in-house R&D (Klevorick et al., 1995; Debackere and Veugelers, 2005). Knowledge from universities is often sourced when firms aim at creating entirely new technology markets so that science can provide a roadmap for industrial research (Tether, 2002; Fleming and Sorensen, 2004). Firms also source patented knowledge from universities that has the power to block competitors (Czarnitzki et al., 2012). In addition, the generic nature of knowledge from universities and public research institutions is less likely to lead to appropriation issues as compared with rather applied knowledge produced for subsequent commercialization (Cassiman and Veugelers, 2002; Czarnitzki et al., 2015).

Among the knowledge sources discussed here, competitors are the most sensitive source of knowledge. Competitors operate similar products and technologies in the same market under the same economic conditions, such as industry regulations, so that rivals' knowledge is most similar to the knowledge of the firm itself and may therefore be most valuable for improving own products, processes, and strategies. Too close ties to product or technology market rivals, however, bear the risk of leakage of important own technological advances, on the one hand, and of disclosing own strengths and weaknesses, on the other hand. This information can strengthen rivals. Accordingly, firms are reluctant to share knowledge with rivals because appropriation is crucial (Czarnitzki et al., 2015).

3.1.2. Heterogeneous knowledge sources through a social identity theory lens

Social identity theory indicates that social comparison is most crucial when it affects similar organizations (Tajfel and Turner, 1986; Abrams and Hogg, 1990; Branscombe et al., 1993). With regard to external knowledge sources, competitors are the most similar type of organization. Competitors can hence be considered the most interesting source of knowledge for the focal company because they have the most relevant knowledge about markets, products, and technologies.

The valuation of competitors' knowledge, technologies, and products by corporate insiders goes hand in hand with a comparison of the focal firm and the competitor along the same dimensions of expertise (Caddick, 1982). The appreciation of competitors' knowledge and technologies, therefore, enforces the acknowledgment of own strengths and weaknesses. Social comparison with competitors can trigger a strong need to differentiate the in-group from the out-group (Sanchez-Mazas et al., 1994). Individuals can react with increased efforts to reassure distinctiveness and to reinstall the boundaries between groups in order to protect their self-concept and the identity and integrity of their company (Jetten and Spears, 2003). Resistance against competitor knowledge or degradation of external knowledge from rivals can occur in response.

With regard to other types of knowledge sources, organizations can make social comparisons along different dimensions and value complementary or different knowledge (Tajfel and Turner, 1986). Comparisons can be made along different dimensions so that the competencies of suppliers, customers, and universities can be acknowledged without threatening the organizational identity of the firm. Along these lines, we hypothesize:

Hypothesis 2: External knowledge sourcing from competitors increases the likelihood of internal resistance against new innovation projects more strongly than external knowledge sourcing from vertical partners and universities.

3.2. *Firm success and the NIH syndrome*

In-group favoritism can also be determined by the extent to which individuals identify with the in-group. If the in-group consists of individuals that show a strong identification with the group, there is a superior tendency towards in-group favoritism and a high willingness to take defensive actions against out-groups (Branscombe et al., 1993; Doosje et al., 1995; Spears et al., 1997). Group members that identify strongly with their in-group are motivated to differentiate themselves from the out-group. In case identification with the group is low, group members may be insufficiently aware of or only marginally interested in group identity so that they do not take actions to preserve group identity (Jetten and Spears, 2003).

The extent to which individuals identify themselves with their company increases with relative group success because success increases the group's distinctiveness and attractiveness (Blanchard et al., 1975; Dutton et al., 1994). The more successful a group is, the more self-esteem and satisfaction its members can derive from social comparison with other organizations and the more individuals are tied to their organization. The willingness to take defensive actions against out-groups is higher for groups of high identifiers (Branscombe et al., 1993; Doosje et al., 1995; Spears et al., 1997) because the valuation of out-groups can evoke the feeling of inferiority *vis-à-vis* out-groups so that the self-esteem of a group members is threatened (Nadler and Fisher, 1986; Nadler, 1991). Hence, we suggest that internal resistance against external knowledge is strongest within successful firms.

Hypothesis 3a: External knowledge sourcing increases the likelihood of internal resistance against new innovation projects more strongly for top performers.

A strong identification with the group is expected to lead to a higher likelihood that external knowledge generated by similar out-groups, i.e. competitors, is rejected because knowledge from such sources constitutes the biggest threat to the group identity. Hence, we hypothesize:

Hypothesis 3b: External knowledge sourcing from competitors increases the likelihood of internal resistance against new innovation projects more strongly for top performers.

4. Data, definition of variables, and descriptive statistics

4.1. *Data*

The empirical analysis is based on the Mannheim Innovation Panel, which is the German part of the Community Innovation Survey (CIS) of the European Commission. The CIS survey of 2003 questionnaire includes questions on internal resistance as a hampering factor for innovation activities. The survey wave 2003 constitutes a cross-sectional database for our empirical analysis. We focus on manufacturing firms only. This leaves us with a sample of 905 firms.

4.2. *Definition of variables and descriptive statistics*

4.2.1. *Dependent variable*

It is difficult to directly observe whether employees have a negative attitude towards externally developed knowledge because there are good reasons to deny a negative attitude if directly asked. In consequence, we use the survey question whether internal resistance led to project delays, cancelations, or terminations in the period 2000–2002 at the firm level. If we

Table 1. Descriptive statistics

#	Full sample 905 Mean (SD)	Less well-performing firms 675	Top performers 230	Mean difference[1]
Internal resistance	0.10	0.11	0.08	0.03
	(0.30)	(0.31)	(0.27)	
External knowledge inflows	0.68	0.67	0.71	–0.04
	(0.47)	(0.47)	(0.45)	
. . . from vertical partners	0.63	0.63	0.62	0.01
	(0.48)	(0.48)	(0.49)	
. . . from competitors	0.24	0.22	0.31	–0.09***
	(0.43)	(0.41)	(0.46)	
. . . from scientific institutions	0.13	0.12	0.17	–0.05**
	(0.34)	(0.33)	(0.38)	
Number of employees	427.94	454.05	351.33	102.72
	(1366.96)	(1484.48)	(939.46)	
Log (employees)	4.57	4.60	4.51	0.09
	(1.65)	(1.66)	(1.63)	
Share of low-skilled workers	81.23	82.16	78.52	3.64**
	(19.97)	(19.83)	(20.17)	
Patent stock	8.83	5.61	18.28	–12.67
	(132.32)	(56.24)	(244.32)	
Patent stock/employees	0.02	0.02	0.02	–0.00
	(0.08)	(0.07)	(0.09)	
East Germany	0.33	0.33	0.33	0.00**
	(0.47)	(0.47)	(0.47)	
Part of a firm group	0.45	0.43	0.50	–0.07***
	(0.50)	(0.50)	(0.50)	
. . . with a foreign head quarter	0.12	0.10	0.19	–0.09
	(0.33)	(0.30)	(0.39)	
Age	31.55	32.30	29.32	2.98
	(35.73)	(36.73)	(32.59)	
Log (age)	3.00	3.01	2.98	0.03
	(0.98)	(1.01)	(0.92)	

[1]This column shows the differences in the means of top-performing and less well-performing firms for the variables of interest. Significance levels of *t*-test for a significant difference in the means are presented with asterisks.
*,**,***indicate 10%, 5% and 1% significance levels.
SD, standard deviation.

find a positive relationship with externally acquired knowledge, we interpret this as evidence for the NIH syndrome.

In total, 47 firms reported that at least 1 of their innovation projects was delayed, 20 stated that at least 1 innovation project was canceled, and 39 reported that at least 1 innovation project was not started due to internal resistance. We define a binary indicator that equals one if one of the innovation obstacles occurred and zero otherwise. In total, 90 firms reported that internal resistance had a negative impact on their innovation activities.[3] Descriptive statistics are presented in Table 1.

4.2.2. *Defining top performers*

The distinction between top-performing and other firms is necessary for testing Hypotheses 3a and 3b. Top-performing firms are distinguished from others according to their return on sales. Because survey

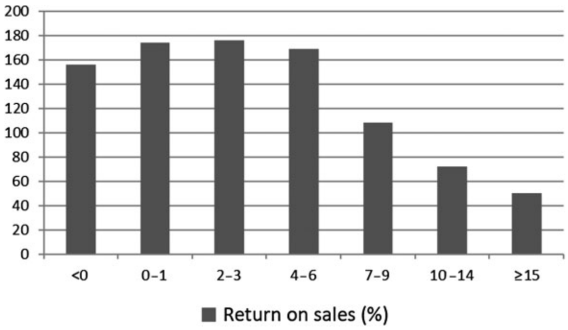

Figure 1. Return on sales distribution.

respondents have been found to be reluctant to report their returns on sales figures, they were given seven different ranges in the survey questionnaire. The distribution of firms across the returns on sales classes is reported in Figure 1. It appears that the number of firms per return on sales class strongly decreases in class (7–10%) and beyond. Hence, we use 7% return on sales as a cut-off point to distinguish strong performers from other sample firms.

4.2.3. Measuring external knowledge inflows

The regressors of main interest capture information about external knowledge inflows. We define a binary variable that equals one if external knowledge was acquired for a process and/or product innovation in the period 2000–2002. The majority of 617 sample firms reported external knowledge inflows. We distinguish between external knowledge acquired from competitors, suppliers, or customers (vertical relationships) and scientific institutions that led to a process and/or product innovation. Most of the firms (569) acquire knowledge from vertical partners. A much smaller share of firms acquires knowledge from scientific institutions (121) and competitors (218).

4.2.4. Control variables

We use a rich set of control variables. First, we use the number of employees as a measure for firm size. The conflict potential and, hence, the likelihood of internal resistance increases with the number of employees as large firms typically have a more complex organizational structure. For the empirical analysis, the logarithm of firms' labor force is used to take account of the skewness of the firm size distribution.

Moreover, we control for firms' innovation activities as measured by the firms' patent stock.[4] We calculate the patent stock as follows:

$$\textit{patent stock}_{it} = (1-.15)\,\textit{patent stock}_{it-1} + \textit{patent applications}_{it}$$

We orthogonalize this variable using the firm size variable. We also control for the human capital composition within a firm's labor force by defining a variable that captures the share of low-skilled workers.

Moreover, firm age is taken into account because organizational complexity is likely to increase with age. We also control for firms being part of a firm group. Lastly, we control for firms' industry affiliation by means of nine industry dummies and for firm location in East Germany.[5]

5. Empirical results

We test our hypotheses using a series of probability models. The dependent variable is a binary variable indicating internal resistance against innovation projects, which we relate to the external knowledge acquisition. The regression results are presented in Tables 2 and 3.

The first column of Table 2 presents the test for the overall presence of the NIH syndrome (Hypothesis 1). The results show that the likelihood of internal resistance is not significantly affected by external knowledge inflows for the average company. Hence, Hypothesis 1 does not receive support.

The second specification in Table 2 distinguishes between different sources of external knowledge by including dummy variables indicating knowledge inflow from vertical partners, competitors, and scientific organizations. It appears that the NIH syndrome only occurs for knowledge inflows from competitors, confirming Hypothesis 2. External knowledge from competitors increases the likelihood of internal resistance increases by 35% at the means of all other variables.[6] This finding is in line with the prediction that resistance is strongest if the out-group is similar to the in-group.

With regard to the control variables, the results show that internal resistance is mainly determined by firm size. The larger the firm, the more complex the organization and the greater the conflict potential. Most other variables do not impact on the likelihood of internal resistance.[7,8]

Table 3 shows the regression results for Hypotheses 3a and 3b. We distinguish between the top-performing firms and firms with a medium or low performance.[9] The results show that there is no evidence for a higher likelihood of internal resistance if firms experience external knowledge inflows from any type of source, neither among the top-performing

Table 2. Probit models for the likelihood of internal resistance

Estimation approach	I probit Coeff. (SE)	II probit
External knowledge inflows	0.21	
	(0.15)	
. . . from vertical partners		0.10
		(0.14)
. . . from competitors		0.27**
		(0.13)
. . . from universities		0.20
		(0.15)
Log (employees)	0.13***	0.12***
	(0.04)	(0.04)
Share of low-skilled workers	−0.01	−0.01
	(0.00)	(0.00)
Patent stock/employees	0.34	0.35
	(0.71)	(0.72)
East Germany	−0.26*	−0.28*
	(0.15)	(0.15)
Part of a firm group	0.09	0.07
	(0.15)	(0.15)
. . . with a foreign head quarter	0.01	0.03
	(0.18)	(0.18)
Log (age)	−0.02	−0.02
	(0.07)	(0.07)
Constant	−1.44***	−1.41***
	(0.43)	(0.43)
N	905	905
Log-likelihood	−276.92	−274.74
Hosmer–Lemeshow χ^2	9.29	8.14
McKelvey and Zavoina R^2	0.10	0.11

Eight industry dummies are included in all specifications.
*,**,***indicate 10%, 5%, and 1% significance levels.
SE, standard error.

firms (column 2) nor among the less well-performing firms (column 1). We, hence, do not find support for Hypothesis 3a.

Columns 3 and 4 show the results distinguishing between different knowledge sources. It appears that the top-performing firms are more likely to experience internal resistance if knowledge is acquired from competitors, whereas there is no such effect for the less well-performing firms. This supports Hypothesis 3b. The finding is in line with the argument that individuals within successful companies identify more strongly with their in-group, so they react defensively if their group identity is threatened by the expertise of similar groups.

Regarding the control variables, we find for the subsample of medium- or low-performing companies that company size matters. For this subsample, the share of low-skilled workers has a significant impact as well. The higher the share of low-skilled workers, the lower is the likelihood of internal resistance. For top performers, we find that being part of a firm group impacts the likelihood of internal resistance significantly in a positive way.[10,11]

6. Concluding remarks

The NIH syndrome is well known among practitioners, but received relatively little attention in the academic literature (Antons and Piller, 2015). This study contributes to our understanding of the antecedents of the NIH syndrome by showing that the source of external knowledge and the success of the company matter.

Drawing from social identity theory and the concept of organizational identity, we argue and show that internal resistance against external knowledge is strongest if the out-group from which the knowledge is acquired is similar to the in-group. If the out-group shares characteristics that are important for in-group identification, individuals fear their group identity threatened and might take defensive actions to reinstall the boundaries between the groups. Moreover, our findings suggest that the NIH syndrome against external knowledge from competitors is more likely to occur within successful companies. The success of a firm generates satisfaction among its insiders. In-group favoritism increases as does the readiness of insiders to reject external knowledge from competitors in order to avoid comparisons along the same dimensions of expertise to protect the distinctiveness and self-esteem of the group and of the individual members (Nadler and Fisher, 1986; Nadler, 1991). Resistance occurs as a means to affirm a positive social identity (Turner et al., 1987).

Our results have important implications for management. We have shown that it is not only individual- and team-related factors or misaligned communication and incentive schemes that facilitate the NIH syndrome as previous literature suggests, but that the source of external knowledge and company success matter as well. Managers should, hence, take the source of external knowledge into account when deciding on their knowledge integration strategies. If the loci of knowledge creation share important characteristics that are supposed to distinguish the in-group from out-groups, special means should be

Table 3. Probit models for the likelihood of internal resistance: high performers versus median and low performers

Sample	I Low and medium performers Return on sales ≤ 7% Coeff. (SE)	II High performers Return on sales > 7%	III Low and medium performers Return on sales ≤ 7%	IV High performers Return on sales > 7%
External knowledge inflows	0.22	0.27		
	(0.17)	(0.37)		
. . . from vertical partners			0.24	–0.67*
			(0.17)	(0.39)
. . . from competitors			0.19	0.79**
			(0.15)	(0.35)
. . . from universities			–0.06	0.52
			(0.18)	(0.36)
Log (employees)	0.16***	–0.03	0.15***	–0.09
	(0.05)	(0.12)	(0.05)	(0.12)
Share of low-skilled workers	–0.01**	0.00	–0.01**	0.00
	(0.00)	(0.01)	(0.00)	(0.01)
Patent stock/employees	0.49	0.29	0.44	0.44
	(0.87)	(1.67)	(0.89)	(1.96)
East Germany	–0.26	–0.45	–0.28*	–0.50
	(0.16)	(0.40)	(0.17)	(0.42)
Part of a firm group	–0.14	1.62***	–0.14	1.75***
	(0.17)	(0.53)	(0.17)	(0.58)
. . . with a foreign head quarter	0.13	–0.35	0.15	–0.37
	(0.23)	(0.33)	(0.23)	(0.37)
Log(age)	–0.04	0.03	–0.04	0.06
	(0.07)	(0.18)	(0.08)	(0.19)
Constant	–1.22***	–3.63***	–1.22***	–3.58***
	(0.47)	(1.17)	(0.47)	(1.17)
N	675	230	675	230
Log-likelihood	–215.33	–48.06	–213.90	–44.17
Hosmer–Lemeshow χ^2	5.34	3.45	4.70	4.01
McKelvey and Zavoina R^2	0.11	0.49	0.12	0.61

Eight industry dummies are included in all specifications.
*,**,***indicate 10%, 5%, and 1% significance levels.
SE, standard error.

put into place in order to support the adaption and integration of external knowledge. Means to prevent the NIH syndrome should be worked out by the management in advance. Especially, the management of successful companies should be aware of their higher risk of experiencing the NIH syndrome. Upper management can take means to influence corporate social identity by stressing distinctive dimensions of competence of the company that are not shared by companies from which knowledge is sourced.

As any, our study is not free of limitations. One limitation relates to the difficulty to observe whether employees have a negative attitude towards externally developed knowledge. Therefore, we use the occurrence of internal resistance at the firm level as a proxy variable. This variable only captures the most significant cases of the NIH syndrome and does not take account of individual negative attitudes that do not have direct consequences. Another limitation arises from the fact that we use the company as the level of analysis. As such, our study has to be seen as complementary to project- or team-level studies.

Acknowledgements

We thank Rene Belderbos, Bruno Cassiman, Maarten Cuijpers, Dirk Czarnitzki, Christoph Grimpe, Hannes Günter, Paul Jensen, and Bart Van Looy for helpful comments. We also thank the participants of the CISS summer school 2011 and the participants of the DRUID 2011 for helpful discussion. This paper was nominated for the best paper award at the DRUID 2011.

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Notes

1. Overall, Katz and Allen (1982) find a curvilinear effect with regard to team tenure.
2. Garud and Karnoe (2001) define path dependence as '*a sequence of events constituting a self-reinforcing process that unfolds into one of several potential states. The specific state that eventually emerges depends on the particular sequence of events that unfold*'.
3. Note that firms can report several consequences of internal resistance (delays, cancelations, not started projects) at the same time.
4. The survey would allow us to use firms' R&D expenditure as a measure for their innovativeness. R&D expenditure could, however, be influenced by our resistance variable. For instance, if an innovation project is canceled due to internal resistance, the R&D expenditure of that firm would be lower by definition.
5. Bivariate correlations are presented in Table A1 in the Appendix.
6. The percentage change is the marginal effect for a discrete change of the dummy variable from zero to one.
7. Likelihood-ratio (LR) tests for the joint significance of the industry dummies cannot reject the null hypothesis that they are not jointly different from zero. The LR test statistics are 4.26 for the first specification of Table 2 and 4.24 for the second specification respectively.
8. The survey further allows distinguishing between knowledge flows related to product and process innovation. If we do this, we find that the results for knowledge inflows from competitors are driven by the product-related innovation inflows. This is in line with our expectations because competitors are defined as rivals on the product market. This is also in line with the presumption that the type of knowledge matters and should receive further attention. Results are available from the authors upon request.
9. We investigated whether there are systematic differences in terms of returns to sales between knowledge sourcing companies and others. Both firm groups have on average 2–4% returns to sales. Because return to sales is measured as a categorical variable, we conducted a χ^2 test. Equality of the distribution for firms sourcing externally and others could not be rejected.
10. If the top-performing firms are treated separately, industry effects matter. LR tests reject that the eight industry dummies are jointly equal to zero at the 5% level of statistical significance (LR = 8.22** for model II; LR = 8.35** for model IV). For the subsample of low and medium well-performing firms, the LR tests on joint significance are not statistically different from zero. The LR statistics are 7.03 for model I and 7.17 for model II.
11. Because the number of firms that experience internal resistance might be considered as small (10% of the sample), we re-estimated all models using rare event logit models (King and Zeng, 2002; Tomz et al., 2003) as robustness checks. The results are very similar to the findings presented here and are available from the authors upon request. Furthermore, our results are robust to a specification that uses interaction terms to test Hypotheses 3a and 3b instead of an empirical approach that relies on a sample split. The results are available from the authors upon request.

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Appendix

Table A1. Bivariate correlations

	Variables	1	2	3	4	5	6	7	8	9	10	11
1	Internal resistance											
2	External knowledge inflows	0.07										
3	. . . from vertical partners	0.06	0.87									
4	. . . from competitors	0.13	0.38	0.22								
5	. . . from scientific institutions	0.10	0.27	0.19	0.13							
6	Log(employment)	0.15	0.10	0.07	0.18	0.09						
7	Share of low-skilled workers	–0.02	–0.11	–0.12	–0.06	–0.21	0.19					
8	Patent stock/employees	0.04	0.06	0.06	0.00	0.08	0.02	–0.09				
9	East Germany	–0.08	0.05	0.05	0.04	0.03	–0.21	–0.18	–0.07			
10	Part of a firm group	0.10	0.05	0.04	0.15	0.07	0.49	0.02	0.08	–0.13		
11	. . . with a foreign head quarter	0.04	0.00	–0.02	0.01	0.03	0.21	–0.01	0.02	–0.07	0.42	
12	Log(age)	0.04	0.01	–0.02	0.02	–0.05	0.32	0.25	0.04	–0.37	0.06	–0.01